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Studierichting: Informatica
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$$\int_0^{\frac{\pi}{4}} \tan x dx$$

$$\int \tan x dx$$

$$= \int \frac{\sin x}{\cos x} dx$$

Substitutie: $t = \cos x \Rightarrow dt = -\sin x dx$

$$= -\int \frac{1}{t} dt$$

$$= -\ln|\cos x| + c$$

$$\Rightarrow \int_0^{\frac{\pi}{4}} \tan x dx = -\ln\left|\cos\frac{\pi}{4}\right| + \ln|\cos 0| = -\ln\frac{\sqrt{2}}{2} + \ln 1 = -\ln\frac{\sqrt{2}}{2}$$

References